Radon Measures in the Spirit of Bourbaki

Throughout, X is a locally compact Hausdorff space. We write $C_c(X;\mathbb{R})$ for the space of real-valued continuous functions with compact support, and $C_c^+(X)$ for its nonnegative cone. All inequalities between functions are pointwise. Nonnegative functions may take the value $+\infty$; we use $0 \cdot (+\infty) = 0$.

Definition 0.1. A positive Radon measure, in the functional sense, is a linear map $\mu : C_c(X;\mathbb{R}) \to \mathbb{R}$ such that $\mu(\varphi) \geq 0$ for every $\varphi \in C_c^+(X)$.

We fix such a functional μ. Positivity implies monotonicity: if $\varphi \leq \psi$, then $\mu(\varphi) \leq \mu(\psi)$.

Lemma 0.2 (Compact cutoffs)**.** *If $K \subseteq U \subseteq X$, with K compact and U open, there exist an open set W and $\eta \in C_c^+(X)$ such that*

$$K \subseteq W \subseteq \overline{W} \subseteq U, \qquad \overline{W} \text{ is compact}, \qquad 0 \leq \eta \leq 1, \quad \eta|_K = 1, \quad \operatorname{supp}\eta \subseteq U.$$

Proof. Local compactness and the Hausdorff property give, for each $x \in U$, an open neighbourhood whose compact closure lies in U. Indeed, first take a compact neighbourhood and then use regularity of that compact Hausdorff subspace to shrink inside U. A finite covering of K by such neighbourhoods gives W. Applying the same construction to $\overline{W} \subseteq U$, choose an open V with compact closure and $\overline{W} \subseteq V \subseteq \overline{V} \subseteq U$. In the compact Hausdorff space $\overline{V}$, Urysohn's lemma gives a continuous function equal to 1 on K and 0 on $\overline{V} \setminus W$, with values in $[0,1]$. Extend it by zero outside $\overline{V}$. The resulting function is continuous, and its support is contained in $\overline{W}$. □

1. Lower semicontinuous functions

Definition 1.1. A function $u : X \to [-\infty, +\infty]$ is lower semicontinuous if $\{x \in X : u(x) > a\}$ is open for every $a \in \mathbb{R}$. We write $I^+(X)$ for the set of nonnegative lower semicontinuous functions $u : X \to [0, +\infty]$.

Proposition 1.2. *The sum of two elements of $I^+(X)$, the pointwise supremum of any nonempty family in $I^+(X)$, and the sum of any sequence in $I^+(X)$ belong to $I^+(X)$.*

Proof. For $u, v \in I^+(X)$ and $a \in \mathbb{R}$,

$$\{u + v > a\} = \bigcup_{r \in \mathbb{Q}} \big(\{u > r\} \cap \{v > a - r\}\big).$$

If $u = \sup_{h \in H} h$, then $\{u > a\} = \bigcup_{h \in H} \{h > a\}$. Finally, a nonnegative series is the supremum of its finite partial sums. □

Definition 1.3. A nonempty family H of functions is upward directed if, for every $a, b \in H$, there exists $c \in H$ with $a \leq c$ and $b \leq c$. For $u \in I^+(X)$, put

$$D_u = \{\varphi \in C_c^+(X) : \varphi \leq u\}.$$

Proposition 1.4. *For every $u \in I^+(X)$, the family D_u is upward directed and $u = \sup_{\varphi \in D_u} \varphi$.*

Proof. The zero function belongs to D_u, and $\max(\varphi, \psi) \in D_u$ whenever $\varphi, \psi \in D_u$. Fix $x \in X$ and a finite number t with $0 < t < u(x)$. The set $U = \{u > t\}$ is open. Lemma 0.2 gives $\eta \in C_c^+(X)$ with $\eta(x) = 1$, $0 \leq \eta \leq 1$, and $\operatorname{supp}\eta \subseteq U$. Then $t\eta \in D_u$, so $\sup_{\varphi \in D_u} \varphi(x) \geq t$. Letting $t \uparrow u(x)$ proves the assertion when $u(x) > 0$, including $u(x) = +\infty$. The case $u(x) = 0$ is immediate. □

Definition 1.5. For $u \in I^+(X)$, define

$$J(u) = \sup\{\mu(\varphi) : \varphi \in D_u\} \in [0, +\infty].$$

Proposition 1.6. *The map J is monotone and positively homogeneous. Moreover, $J(\varphi) = \mu(\varphi)$ for every $\varphi \in C_c^+(X)$.*

Proof. If $u \leq v$, then $D_u \subseteq D_v$. For $c > 0$, $D_{cu} = cD_u$, which proves homogeneity; the case $c = 0$ follows from $J(0) = 0$. Finally, $\varphi \in D_\varphi$ and every $\psi \in D_\varphi$ satisfies $\mu(\psi) \leq \mu(\varphi)$. □

Theorem 1.7 (Directed monotone convergence)**.** *Let $H \subseteq I^+(X)$ be upward directed, and let $u = \sup_{h\in H} h$. Then $u \in I^+(X)$ and*

$$J(u) = \sup_{h\in H} J(h).$$

Proof. Proposition 1.2 gives $u \in I^+(X)$, and monotonicity gives $J(u) \geq \sup_{h\in H} J(h)$. To prove the reverse inequality, fix $\varphi \in D_u$ and $\varepsilon > 0$. Set $\psi = (\varphi - \varepsilon)^+$ and $K = \operatorname{supp}\psi$. Then K is compact and $K \subseteq \{\varphi \geq \varepsilon\}$. In particular, $\psi(x) < \varphi(x) \leq u(x)$ for every $x \in K$.

For each $x \in K$, choose $h_x \in H$ such that $h_x(x) > \psi(x)$. The function $h_x - \psi$ is lower semicontinuous, so $O_x = \{h_x > \psi\}$ is an open neighbourhood of x. Here subtraction is well defined since ψ is finite; lower semicontinuity follows, for example, by writing

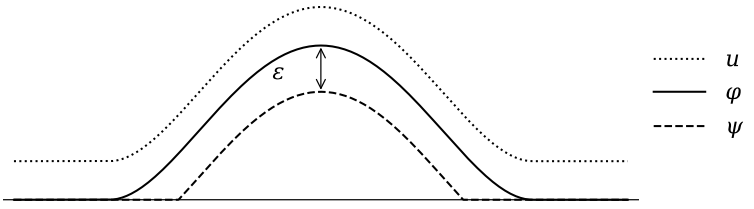
$$\{h_x - \psi > a\} = \bigcup_{r\in\mathbb{Q}} \big(\{h_x > r\} \cap \{\psi < r - a\}\big).$$

Compactness gives a finite subcover of K, and directedness gives one $h \in H$ dominating the corresponding finitely many h_x. Thus $\psi \leq h$ on K, and also on $X \setminus K$ because $\psi = 0$ there. If $K = \varnothing$, any $h \in H$ suffices.

Choose a cutoff $\eta \in C_c^+(X)$ equal to 1 on $\operatorname{supp}\varphi$. Since $0 \leq \varphi - \psi \leq \varepsilon\eta$,

$$\mu(\varphi) \leq \mu(\psi) + \varepsilon\mu(\eta) \leq \sup_{h\in H} J(h) + \varepsilon\mu(\eta).$$

Let $\varepsilon \downarrow 0$, keeping η fixed, and then take the supremum over $\varphi \in D_u$. □



The truncation $\psi = (\varphi - \varepsilon)^+$ satisfies $\operatorname{supp}\psi \subseteq \{\varphi \geq \varepsilon\}$ and $0 \leq \varphi - \psi \leq \varepsilon$.

Theorem 1.8 (Additivity of J)**.** *For every sequence $(u_n)_{n\geq 1}$ in $I^+(X)$,*

$$J\left(\sum_{n=1}^{\infty} u_n\right) = \sum_{n=1}^{\infty} J(u_n).$$

Proof. First take $u, v \in I^+(X)$. The family $\{\varphi + \psi : \varphi \in D_u,\ \psi \in D_v\}$ is upward directed and has pointwise supremum $u + v$, by Proposition 1.4. Theorem 1.7 and linearity of μ give

$$\begin{aligned} J(u+v) &= \sup_{\varphi \in D_u,\, \psi \in D_v} \mu(\varphi + \psi) \\ &= \sup_{\varphi \in D_u} \mu(\varphi) + \sup_{\psi \in D_v} \mu(\psi) = J(u) + J(v). \end{aligned}$$

Apply this identity to finite partial sums, and then apply Theorem 1.7 to their increasing sequence. □

2. The upper integral

Definition 2.1. For an arbitrary function $f : X \to [0, +\infty]$, define its upper integral by

$$\mu^*(f) = \inf\{J(u) : u \in I^+(X),\ f \leq u\}.$$

The admissible family is nonempty because it contains the constant function $+\infty$.

Proposition 2.2. *The upper integral is monotone and positively homogeneous, and $\mu^*(u) = J(u)$ for $u \in I^+(X)$. For arbitrary functions $f_n : X \to [0, +\infty]$,*

$$\mu^*\left(\sum_{n=1}^{\infty} f_n\right) \leq \sum_{n=1}^{\infty} \mu^*(f_n).$$

Proof. If $f \leq g$, every lower semicontinuous majorant of g is also one of f. For $c > 0$, multiplication by c is a bijection between the majorants of f and those of cf; homogeneity follows from that of J. Also $\mu^*(0) = 0$. If $u \in I^+(X)$, it is its own majorant, and every majorant v of u satisfies $J(v) \geq J(u)$.

For subadditivity, there is nothing to prove if $\sum_n \mu^*(f_n) = +\infty$. Otherwise, for $\varepsilon > 0$ choose $u_n \in I^+(X)$ such that $f_n \leq u_n$ and $J(u_n) \leq \mu^*(f_n) + \varepsilon 2^{-n}$. By Theorem 1.8,

$$\mu^*\left(\sum_n f_n\right) \leq J\left(\sum_n u_n\right) = \sum_n J(u_n) \leq \sum_n \mu^*(f_n) + \varepsilon.$$

Let $\varepsilon \downarrow 0$. □

3. Recovering a measure on sets

Definition 3.1. For $A \subseteq X$, put $\nu^*(A) = \mu^*(\mathbf{1}_A)$. If U is open, write $m(U) = \nu^*(U)$. Since $\mathbf{1}_U \in I^+(X)$,

$$m(U) = J(\mathbf{1}_U) = \sup\{\mu(\varphi) : \varphi \in C_c^+(X),\ \varphi \leq \mathbf{1}_U\}.$$

Proposition 3.2. *The set function ν^* is an outer measure. It is finite on compact sets and satisfies*

$$\nu^*(A) = \inf\{m(U) : A \subseteq U,\ U \text{ open}\}, \qquad A \subseteq X, \tag{3.1}$$

$$m(U) = \sup\{\nu^*(K) : K \subseteq U,\ K \text{ compact}\}, \qquad U \text{ open}. \tag{3.2}$$

Proof. We have $\nu^*(\varnothing) = 0$, monotonicity follows from Proposition 2.2, and $\mathbf{1}_{\bigcup_n A_n} \leq \sum_n \mathbf{1}_{A_n}$ gives countable subadditivity. If K is compact, a cutoff $\eta \in C_c^+(X)$ equal to 1 on K gives

$$\nu^*(K) \leq \mu^*(\eta) = J(\eta) = \mu(\eta) < +\infty.$$

For (3.1), monotonicity gives the inequality $\nu^*(A) \leq \inf_{U \supseteq A} m(U)$. Conversely, take $f \in I^+(X)$ with $f \geq \mathbf{1}_A$. For $0 < t < 1$, the open set $U_t = \{f > t\}$ contains A, and $t\mathbf{1}_{U_t} \leq f$. Hence

$$\inf_{U \supseteq A,\, U \text{ open}} m(U) \leq m(U_t) \leq t^{-1} J(f).$$

Letting $t \uparrow 1$ and taking the infimum over f proves the reverse inequality. The argument includes infinite values.

For (3.2), let $S = \sup_{K \subseteq U,\, K \text{ compact}} \nu^*(K)$. Monotonicity gives $S \leq m(U)$. Take $\varphi \in C_c^+(X)$ with $\varphi \leq \mathbf{1}_U$ and $\varepsilon > 0$. Then $\psi = (\varphi - \varepsilon)^+$ has compact support $K_\varepsilon \subseteq \{\varphi \geq \varepsilon\} \subseteq U$, and $\psi \leq \mathbf{1}_{K_\varepsilon}$. Thus $\mu(\psi) = \mu^*(\psi) \leq S$. For a fixed cutoff η equal to 1 on $\operatorname{supp} \varphi$, the estimate $0 \leq \varphi - \psi \leq \varepsilon\eta$ gives

$$\mu(\varphi) \leq S + \varepsilon\mu(\eta).$$

Let $\varepsilon \downarrow 0$ and take the supremum over φ to obtain $m(U) \leq S$. □

Let $\mathcal{M}$ be the Carathéodory class of ν^*, namely the sets $E \subseteq X$ such that

$$\nu^*(A) = \nu^*(A \cap E) + \nu^*(A \setminus E) \qquad (A \subseteq X).$$

Carathéodory's theorem states that $\mathcal{M}$ is a σ-algebra and that $\bar{\nu} = \nu^*|_{\mathcal{M}}$ is a complete measure.

Theorem 3.3. *Every Borel set belongs to $\mathcal{M}$. The Borel measure $\nu = \bar{\nu}|_{\mathcal{B}(X)}$ is finite on compact sets, outer regular on Borel sets, and inner regular on open sets. Moreover, $\bar{\nu}$ is outer regular on $\mathcal{M}$.*

Proof. It suffices to show that every open set U belongs to $\mathcal{M}$. For an arbitrary $A \subseteq X$, subadditivity gives one inequality in the Carathéodory identity. If $\nu^*(A) = +\infty$, the reverse inequality is automatic. Suppose therefore that $\nu^*(A) < +\infty$, and fix $\varepsilon > 0$. By (3.1), choose an open $W \supseteq A$ with $m(W) \leq \nu^*(A) + \varepsilon$. By (3.2), choose a compact $K \subseteq W \cap U$ such that

$$m(W \cap U) \leq \nu^*(K) + \varepsilon.$$

Choose an open V with $K \subseteq V \subseteq \overline{V} \subseteq W \cap U$ and compact closure. The disjoint open sets V and $W \setminus \overline{V}$ lie in W. Additivity and monotonicity of J yield

$$\nu^*(K) + m(W \setminus \overline{V}) \leq m(V) + m(W \setminus \overline{V}) \leq m(W).$$

Since $A \cap U \subseteq W \cap U$ and $A \setminus U \subseteq W \setminus \overline{V}$, it follows that

$$\begin{aligned} \nu^*(A \cap U) + \nu^*(A \setminus U) &\leq m(W \cap U) + m(W \setminus \overline{V}) \\ &\leq m(W) + \varepsilon \leq \nu^*(A) + 2\varepsilon. \end{aligned}$$

Let $\varepsilon \downarrow 0$. Thus $U \in \mathcal{M}$, and $\mathcal{B}(X) \subseteq \mathcal{M}$. The remaining assertions follow from Proposition 3.2, since open and compact sets are Borel. In particular, (3.1) applies to every $A \in \mathcal{M}$. □

4. Representation by the Lebesgue integral

We now use the Lebesgue integral associated with the Borel measure ν. Simple functions are understood to be finite-valued. In particular, its value on a nonnegative Borel simple function $\sum_i a_i \mathbf{1}_{A_i}$ with disjoint A_i is $\sum_i a_i \nu(A_i)$.

Lemma 4.1. *If K is compact and s is a nonnegative Borel simple function vanishing outside K, then*

$$\mu^*(s) = \int_X s \, \mathrm{d}\nu.$$

Proof. Write $s = \sum_{i=1}^m a_i \mathbf{1}_{A_i}$, where $a_i \geq 0$ and the Borel sets A_i form a partition of K. Subadditivity and homogeneity give

$$\mu^*(s) \leq \sum_{i=1}^m a_i \mu^*(\mathbf{1}_{A_i}) = \sum_{i=1}^m a_i \nu(A_i) = \int_X s \, \mathrm{d}\nu.$$

Choose a finite $M \geq \max_i a_i$ and apply the same inequality to $M\mathbf{1}_K - s$. Then

$$\begin{aligned} M\nu(K) = \mu^*(M\mathbf{1}_K) &\leq \mu^*(s) + \mu^*(M\mathbf{1}_K - s) \\ &\leq \mu^*(s) + M\nu(K) - \int_X s \, \mathrm{d}\nu. \end{aligned}$$

Every term is finite, since $\nu(K) < +\infty$ and s is bounded. Cancelling $M\nu(K)$ gives the reverse inequality. □

Proposition 4.2. *For every $f \in C_c(X; \mathbb{R})$, the function f is ν-integrable and*

$$\mu(f) = \int_X f \, \mathrm{d}\nu.$$

Proof. First suppose $f \geq 0$ and put $K = \operatorname{supp} f$. The measure of K is finite, so f is integrable. The case $f = 0$ is immediate. For $\varepsilon > 0$, the function $s = \varepsilon \lfloor f/\varepsilon \rfloor$ is a nonnegative Borel simple function vanishing outside K, and

$$s \leq f \leq s + \varepsilon \mathbf{1}_K.$$

Lemma 4.1, monotonicity and subadditivity give

$$\int_X s \, \mathrm{d}\nu \leq \mu^*(f) \leq \int_X s \, \mathrm{d}\nu + \varepsilon\nu(K).$$

Monotonicity of the Lebesgue integral gives the same bounds for $\int_X f \, \mathrm{d}\nu$. Hence

$$\left| \mu^*(f) - \int_X f \, \mathrm{d}\nu \right| \leq \varepsilon\nu(K).$$

Let $\varepsilon \downarrow 0$. Since $\mu(f) = J(f) = \mu^*(f)$, the desired formula follows. For real-valued f, both f^+ and f^- belong to $C_c^+(X)$. Apply the result to them and use linearity. □

Proposition 4.3 (Uniqueness)**.** *Suppose ρ is a positive Borel measure, inner regular on open sets and outer regular on Borel sets, such that every $f \in C_c(X;\mathbb{R})$ is ρ-integrable and $\mu(f) = \int_X f \, \mathrm{d}\rho$. Then $\rho = \nu$.*

Proof. For every open set U,

$$\nu(U) = J(\mathbf{1}_U) = \sup_{\substack{g \in C_c^+(X) \\ g \le \mathbf{1}_U}} \mu(g).$$

For each admissible g, $\mu(g) = \int_X g \, \mathrm{d}\rho \le \rho(U)$. Thus $\nu(U) \le \rho(U)$. Conversely, for a compact $K \subseteq U$, Lemma 0.2 gives $g \in C_c^+(X)$ with $\mathbf{1}_K \le g \le \mathbf{1}_U$. Consequently,

$$\rho(K) \le \int_X g \, \mathrm{d}\rho = \mu(g) \le \nu(U).$$

Inner regularity gives $\rho(U) \le \nu(U)$, so the measures agree on open sets. For $A \in \mathcal{B}(X)$, outer regularity now gives

$$\rho(A) = \inf_{\substack{U \supseteq A \\ U \text{ open}}} \rho(U) = \inf_{\substack{U \supseteq A \\ U \text{ open}}} \nu(U) = \nu(A).$$

□

Theorem 4.4 (Riesz–Markov representation theorem)**.** *Let X be a locally compact Hausdorff space, and let $\mu : C_c(X;\mathbb{R}) \to \mathbb{R}$ be a positive linear functional. There exists a unique positive Borel measure ν satisfying all of the following conditions:*

(i) *Every $f \in C_c(X;\mathbb{R})$ is ν-integrable and $\mu(f) = \int_X f \, \mathrm{d}\nu$.*

(ii) *$\nu(K) < +\infty$ for every compact $K \subseteq X$.*

(iii) *For every open $U \subseteq X$,*

$$\nu(U) = \sup\{\nu(K) : K \subseteq U,\ K \text{ compact}\}.$$

(iv) *For every $A \in \mathcal{B}(X)$,*

$$\nu(A) = \inf\{\nu(U) : A \subseteq U,\ U \text{ open}\}.$$

The measure is given by $\nu(A) = \mu^*(\mathbf{1}_A)$ on Borel sets.

Proof. Existence follows from Theorem 3.3 and Proposition 4.2; uniqueness is Proposition 4.3. □